Kärsa Node configuration list

1. Address space (in hexadecimal, links to initialization)

		03	04	05	06	07		09	0A			0D	0E		10
11	12	13	14	15	16										

2. Type parameters

0, KRS_NODE_TYPE_MFC

Parameter	Index	Subinde	Datatype
Settable parameters			
Flow setpoint	0x2F00	0x01	Real32
Readable parameters			
Flow setpoint	0x2F00	0x01	Real32
Flow monitor value	0x2C00	0x01	Real32
Medium temperature	0x2503	0x01	Unsigned16
Device status, temperature	0x2004	0x02	Real32
Device status, voltage	0x2004	0x03	Real32

1, KRS_NODE_TYPE_DIO

Parameter	Index	Subinde	Datatyp
Settable parameters			
Channel output state	0x6200	0x01	Unsigned8
Readable parameters			
Channel output state	0x6200	0x01	Unsigned8
Channel state	0x6000	0x01	Unsigned8

3. Configuration list

NOTE: least significant byte first in data fields

3.1. MFC configurations

3.1.1. MFC configuration A

Device type:	**Bürkert 8742, 100mlpm vacuum**		
MION1 source exhaust	**Bürkert code: 314859**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (100.0%)	0x2F11	0x0A	0x00 00 C8 42
Set 'actual flow' unit 0x823 (Smlpm)	0x2C00	0x02	0x23 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 100.0%	0x2F00	0x01	0x00 00 C8 42

3.1.2. MFC configuration B

Device type:	**Bürkert 8742, 100mlpm pressure**		
MION1 source feed	**Bürkert code: 326683**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (0.0%)	0x2F11	0x0A	0x00 00 00 00
Set 'actual flow' unit 0x823 (Smlpm)	0x2C00	0x02	0x23 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 0.0%	0x2F00	0x01	0x00 00 00 00

3.1.3. MFC configuration C

Device type:	**Bürkert 8742, 50lpm vacuum**		
MION1 main flow	**Bürkert code: 314860**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (0.0%)	0x2F11	0x0A	0x00 00 00 00
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 0.0%	0x2F00	0x01	0x00 00 00 00

3.1.4. MFC configuration D

Device type:	**Bürkert 8742, 5lpm pressure**		
Calibrator sample flow	**Bürkert code: 337354**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (40.0%)	0x2F11	0x0A	0x00 00 20 42
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 40.0%	0x2F00	0x01	0x00 00 20 42

3.1.5. MFC configuration E

Device type:	**Bürkert 8742, 5lpm pressure**		
Flush plate counterflow, SH sample	**Bürkert code: 337354**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (20.0%)	0x2F11	0x0A	0x00 00 A0 41
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 20.0%	0x2F00	0x01	0x00 00 A0 41

3.1.6. MFC configuration F

Device type:	**Bürkert 8742, 50lpm pressure**		
SH sheath 2	**Bürkert code: 363896**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (56.0%)	0x2F11	0x0A	0x00 00 60 42
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 56.0%	0x2F00	0x01	0x00 00 60 42

3.1.7. MFC configuration G

Device type:	**Bürkert 8742, 50lpm vacuum**		
SH main flow	**Bürkert code: 314860**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (55.0%)	0x2F11	0x0A	0x00 00 5C 42
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 55.0%	0x2F00	0x01	0x00 00 5C 42

3.1.8. MFC configuration H

Device type:	**Bürkert 8742, 5lpm pressure**		
SH sheath 1	**Bürkert code: 337354**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (30.0%)	0x2F11	0x0A	0x00 00 F0 41
Set 'actual flow' unit 0x805 (Slpm)	0x2C00	0x02	0x05 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 30.0%	0x2F00	0x01	0x00 00 F0 41

3.1.9. MFC configuration I

Device type:	**Bürkert 8742, 100mlpm pressure**		
SH reagent	**Bürkert code: 363897**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (20.0%)	0x2F11	0x0A	0x00 00 A0 41
Set 'actual flow' unit 0x823 (Smlpm)	0x2C00	0x02	0x23 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 20.0%	0x2F00	0x01	0x00 00 A0 41

3.1.9. MFC configuration J

Device type:	**Bürkert 8742, 100mlpm vacuum**		
MION2 source exhaust	**Bürkert code: 363897**		
Description	Index	Subindex	Data
Set SP source to 0 (Automatic)	0x2F11	0x03	0x00 00
Set Manual SP to 0	0x2F11	0x0D	0x00 00 00 00
Set Communication Loss Behaviour 1 (Use last SP)	0x2F11	0x0B	0x01
Set Startup SP Percent (50.0%)	0x2F11	0x0A	0x00 00 48 42
Set 'actual flow' unit 0x823 (Smlpm)	0x2C00	0x02	0x23 08
Disable TPDO 1	0x1800	0x01	0x00 00 00 80
Disable TPDO 2	0x1801	0x01	0x00 00 00 80
Disable TPDO 4	0x1803	0x01	0x00 00 00 80
Set TPDO 3 (Actual flow) interval to 1000ms	0x1802	0x05	0xE8 03
Set SP to 50.0%	0x2F00	0x01	0x00 00 48 42

3.2. DIO and AI configurations

3.2.1. AI configuration K

Device type:	**MicroControl 6ch AI**		
MION1 / SH DIO	**µCAN.6.ai-SNAP**		
Description	Index	Subindex	Data
Set filter type to 0 (no filter, cycle through channels)	0x61A0	0x01…6	0x00
Set sensor type +-10V	0x6110	0x01…6	0x29 00
Configure PDO 1 to use event driven transmission	0x1800	0x02	0xFF
Configure PDO 2 to use event driven transmission	0x1801	0x02	0xFF
Disable PDO 3 & 4			
Enable channels 1…6	0x6110	0x01…6	0x01

3.2.2. DIO configuration L

Device type:	**MicroControl 8ch Digital I/O**		
MION1 / SH DIO	**µCAN.8.dio-SNAP**		
Description	Index	Subindex	Data
Set port direction 1…4 input 5…8 output	0x5FF5	0x00	0xF0
Disable pdos for output	0x6005	0x00	0x0F
Configure PDO 1 to use event driven transmission	0x1800	0x02	0xFF
Set PDO 1 timer to 5000 (ms)	0x1800	0x05	0x88 13
Enable pdos for input channels	0x6006	0x01	0xF0
Select absolute input level	0x5FF2	0x00	0x00
Set trigger level to 4.4V	0x5FF0	0x00	0x44

4. Node list

CAN ID	**Type**	**Type description**	**Config**	**Physical device**	**Description**
MION1 NODES					
0x03	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 314859	MION MFC1, Source 1 exhaust, vacuum, 100mlpm
0x04	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION MFC2, Source 1 carrier, pressure, 100mlpm
0x05	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 314859	MION MFC3, Source 2 exhaust, vacuum, 100mlpm
0x06	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION MFC4, Source 2 carrier, pressure, 100mlpm
0x07	0	KRS_NODE_TYPE_MFC	C	Bürkert type 8742, configuration code 314860	MION MFC5, Main flow, vacuum, 50lpm

0x09	1	KRS_NODE_TYPE_DIO	L	MicroControl µCAN.8.dio-SNAP	MION digital I/O, 4ch in, 4ch out
0x0A	2	KRS_NODE_TYPE_AI	K	MicroControl µCAN.6.ai-SNAP	MION analog in, 6ch in
MISC DEVICES					
0x0D	0	KRS_NODE_TYPE_MFC	D	Bürkert type 8742, configuration code 337354	Calibrator sample flow, pressure, 5lpm
0x0E	0	KRS_NODE_TYPE_MFC	E	Bürkert type 8742, configuration code 337354	Flush plate counterflow, pressure, 5lpm
SCENTHOUND NODES					
0x10	0	KRS_NODE_TYPE_MFC	F	Bürkert type 8742, configuration code 363896	SH MFC1, Sheath 2, pressure, 50lpm
0x11	0	KRS_NODE_TYPE_MFC	G	Bürkert type 8742, configuration code 314860	SH MFC2, Exhaust, vacuum, 50lpm
0x12	0	KRS_NODE_TYPE_MFC	E	Bürkert type 8742, configuration code 337354	SH MFC3, Sample, pressure, 5lpm
0x13	0	KRS_NODE_TYPE_MFC	H	Bürkert type 8742, configuration code 337354	SH MFC4, Sheath 1, pressure, 5lpm
0x14	0	KRS_NODE_TYPE_MFC	I	Bürkert type 8742, configuration code 363897	SH MFC5, reagent, pressure, 100mlpm
0x15	1	KRS_NODE_TYPE_DIO	L	MicroControl µCAN.8.dio-SNAP	SH digital I/O, 4ch in, 4ch out
0x16	2	KRS_NODE_TYPE_AI	K	MicroControl µCAN.6.ai-SNAP	SH analog in, 6ch in
MION2 CORE NODES					
0x17	0	KRS_NODE_TYPE_MFC	C	Bürkert type 8742, configuration code 314860	MION2 MFC1, Main flow, vacuum, 50lpm
0x18	1	KRS_NODE_TYPE_DIO		MicroControl µCAN.8.dio-SNAP	MION2 xray on input
0x19	1	KRS_NODE_TYPE_DIO		MicroControl µCAN.8.dio-SNAP	MION2 xray alert input
0x1A	1	KRS_NODE_TYPE_DIO		MicroControl µCAN.8.dio-SNAP	MION2 xray on/off control
0x1B	2	KRS_NODE_TYPE_AI	K	MicroControl µCAN.6.ai-SNAP	MION analog in, 6ch in
MION2 SOURCE FLOW NODES					
0x1C	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC1, purge
0x1D	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC1, reagent
0x1E	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC1, exhaust
0x1F	3			Bürkert valve	MION2, SRC1, reagent valve
0x20	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC2, purge
0x21	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC2, reagent

0x22	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC2, exhaust
0x23	3			Bürkert valve	MION2, SRC2, reagent valve
0x24	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC3, purge
0x25	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC3, reagent
0x26	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC3, exhaust
0x27	3			Bürkert valve	MION2, SRC3, reagent valve
0x28	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC4, purge
0x29	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC4, reagent
0x2A	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC4, exhaust
0x2B	3			Bürkert valve	MION2, SRC4, reagent valve
0x2C	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC5, purge
0x2D	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC5, reagent
0x2E	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC5, exhaust
0x2F	3			Bürkert valve	MION2, SRC1, reagent valve
0x30	0	KRS_NODE_TYPE_MFC	A	Bürkert type 8742, configuration code 326683	MION2, SRC6, purge
0x31	0	KRS_NODE_TYPE_MFC	B	Bürkert type 8742, configuration code 326683	MION2, SRC6, reagent
0x32	0	KRS_NODE_TYPE_MFC	J	Bürkert type 8742, configuration code 314859	MION2, SRC6, exhaust
0x33	3			Bürkert valve	MION2, SRC6, reagent valve
MION2 VOLTAGE AND ESI NODES					
0x34	4			Kärsa HV source	SRC1 HV
0x35	4			Kärsa HV source	SRC2 HV
0x36	4			Kärsa HV source	SRC3 HV
0x37	4			Kärsa HV source	SRC4 HV
0x38	4			Kärsa HV source	SRC5 HV
0x39	4			Kärsa HV source	SRC6 HV
0x3A	5			Bürkert 8763	SRC1 ESI pressure
0x3B	5			Bürkert 8763	SRC2 ESI pressure
0x3C	5			Bürkert 8763	SRC3 ESI pressure
0x3D	5			Bürkert 8763	SRC4 ESI pressure
0x3E	5			Bürkert 8763	SRC5 ESI pressure
0x3F	5			Bürkert 8763	SRC6 ESI pressure

5. Device specific data

0x03

Initialization parameters

Parameter	Index	Subindex	Value	Description
Human readable	0xNNNN	0xNN	0x12	Set unit of measurement to gallons/mth

0x09, MION digital I/O, 4ch in, 4ch out

Initialization parameters

Parameter	Index	Subindex	Value	Description
Human readable	0xNNNN	0xNN	0x12	Set unit of measurement to gallons/mth

Physical configuration

- CAN bitrate 1Mbit/s
- CAN address 0x09
- Termination OFF

Channel #	
1	IN, alert from xray head. High means a fault in head.
2	IN, xray enabled. High means that xray is ready to operate. Should be weakly pulled to GND.
3	IN, xray interlock. High means that interlock circuit is closed and xray can be operated. Opening interlock circuit will also set ch2 to low. Should be weakly pulled to GND.
4	IN, xray active. High means that there is current draw on xray head and hence radiation is on.
5	OUT, xray on. High output asks xray head to start producing radiation.
6	OUT, ion filter HV supply on. High output asks ion filter voltage supply to activate
7	Not used
8	Not used

0x0A, MION analog in, 6ch in

Initialization parameters

Parameter	Index	Subindex	Value	Description
Human readable	0xNNNN	0xNN	0x12	Set unit of measurement to gallons/mth

Physical configuration

- CAN bitrate 1Mbit/s
- CAN address 0x0A
- Termination OFF

Channel #	Description		Unit
1	RH	RH=100*Umeas(V)	%
2	T	T=100*Umeas(V)-40	°C
3	P	P=0,4*Umeas(V)	Bar
4	Not used		
5	Ion filter -, approximately 1/50 of actual output	U=k*Umeas(V)+b	V
6	Ion filter -, approximately 1/50 of actual output	U=k*Umeas(V)+b	V

Default k=101, b=0, but settable by user